

■ ESP8266 Fire Detection Video Alert System

1. Introduction

This project uses an LM393 flame sensor with ESP8266 to detect fire and trigger a video alert over WiFi.

2. Components Required

Component	Quantity
ESP8266 (NodeMCU)	1
LM393 Flame Sensor	1
Jumper Wires	Few
Breadboard	1

3. Wiring Connections

Connect LM393 sensor to ESP8266 as follows:

Sensor Pin	ESP8266 Pin
VCC	3.3V
GND	GND
DO	D5 (GPIO14)

4. Source Code

(Upload this using Arduino IDE) - Connect to WiFi - Detect flame - Serve webpage with video alert - Hold detection for 20 seconds

5. Working Principle

When flame is detected, ESP8266 triggers a web interface. The video is preloaded and plays instantly. The system holds detection for 20 seconds to avoid flickering.

6. Conclusion

This project demonstrates IoT-based fire detection with real-time video alert.